Developing a Model to Adjust Robot Response using Touch

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***Abstract*—Touch is an interesting and alternative method of interaction between robots and humans. There are several areas of application, including medical and manufacturing fields, where human-robot collaboration is important for safety and efficiency. Although there is high potential for this type of interface in industry, there are several research areas in this domain that need to be explored for widespread adoption. These include the robot sensing touch, understanding touch intent, and then adjusting the robot's movements to fit the user's preferences. The robot's reaction to the touch is important because the behaviour directly impacts the interest of the participant to use and engage. In this article, we propose a method of embedding trust into a touch-based interaction where the robot will adjust its state and speed based on the user's trust preferences and touch interactions. We discuss the model development, how it was implemented in the simulation environment, the sample user study to test the system, and the future work. The goal of this work is to develop a model that can combine trust with touch to apply to different domains.**

***Index Terms*—Touch, Trust, Collaborative Robots, Modelling, Proprioceptive Sensing**

I. Introduction

Human–robot collaboration has advanced rapidly through the use of technological tools and techniques designed to improve safety, awareness, and efficiency [1], [2]. Sensors, cameras, and vision-based systems allow robots to perceive their surroundings, detect human presence, and adapt their actions in real time, easing human-robot collaboration in different settings [1].

While there has been a heavy focus on using technology to understand the environment in which human-robot collaboration takes place, other forms of communication in these collaboration tasks have been less explored [1]. In particular, the use of touch is an area that has room for growth and development, especially as a form of communication in collaboration settings between humans and robots. There is an opportunity to use touch to enhance communication, improve safety, and the quality of the interactions.

A. Motivation

Touch is an innovative alternative interface for robots to receive information about their environment through contact. Humans often use touch as a method of communication in human-to-human interaction, making touch an interesting area of exploration for human-robot interactions. However, there are still several issues that exist when using touch as an interface [3]. First, reliably detecting contact using only proprioceptive sensing remains challenging. Although a variety of tactile skins and sensing technologies have been developed, there is still no accurate way of measuring touch in a natural manner [1]. In addition, a touch can be of varying duration and magnitude, making it difficult to understand the intent of a touch, as it can vary depending on the individual. Assuming appropriate identification and classification of the touch and understanding the reaction the robot should have is still under research.

The reaction of the robot to the touch is important because its behaviour will directly influence the interest in using and engaging with the human who is interacting [4], [5]. There are several factors that can impact the trust that a human may have in the robot (i.e., appearance, speed, etc.). In order to ensure participants feel comfortable using the robot, it is important to account for trust in the reaction of the robot. Various interaction parameters derived from human contact prior to and during the event can be used to dictate an appropriate response of the robot arm [6]–[8]. In addition, identifying appropriate movements of the robotic arm should be devised based on the safety limitations of the robot. There is an opportunity to understand how various inputs can be used to dictate how the robotic arm will react to intentional human touch.

B. Application

Although there are several issues with using touch as an interface for human-robot collaboration, there are many areas of application where this may be used. One area is the medical field, where a professional in the field may use touch to adjust a robotic arm during a procedure [9]. In the manufacturing domain, if a robot needs to complete a task or explore its surroundings, touch can be used as a way to guide the arm or to collaborate with the human to complete a task [10]. In order for robots to become more integrated into workplaces and various environments, they must support safe and intuitive interactions with humans [10].

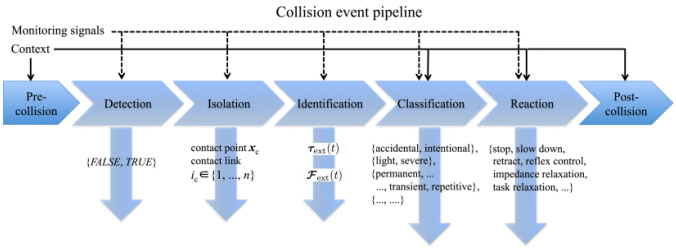

Fig. 1. The collision event pipeline [3]

C. Problem Statement

Proprioceptive touch sensing provides an excellent opportunity for robots to have a new way of sensing their environment that comes directly from humans. However, the relationship between touch and the resulting human-robot interaction is important to ensure that the technology is accepted within the field. A key challenge in this area is trust. The goal of this project is to develop a framework that embeds trust into robot behaviour through touch-based interactions. By enabling robots to respond to physical contact in ways that reflect intent, responsiveness, and reliability, the framework aims to create more natural, predictable, and trustworthy human–robot collaboration. It will enable a more intuitive way for humans to interact with robots and further support human-robot collaboration in the future.

II. Background

A. Using Touch for Human Robot Collaboration

Touch has been researched quite extensively in human-robot collaboration. In this context, the use of touch has been used for starting and stopping a robot [11], [12] or within the collision context [3], [13]. Haddad et al. (2017) define the collision event pipeline when using touch in Figure 1. There are several areas in this pipeline that need further exploration. For example, isolating and identifying a touch is still challenging due to the current technological capabilities and the varying intents that can exist behind a touch and the way it's applied onto the robot. This then leads to difficulties with classification and understanding what a specific touch means in a use case, and how that can result in a change in the reaction of the robot. There are several articles which have explored elements of this collision pipeline. Kim et al. (2021) researched a new sensing method for robot manipulators [14]. Heo et al. (2019) explored how collisions could be detected in the industrial setting [10]. Gamboa-Montero et al. (2022) proposed a system to localise contact on the service robot [15]. Jung et al. (2025) have started exploring how intent can be communicated using common hand gestures to the robot using touch [16]. Although there has been some work combining contact detection with understanding intent, there have been few that have also incorporated trust factors into the analysis.

B. Trust in Human Robot Collaboration

Trust in human-robot collaboration has been extensively explored with multiple factors that may impact the amount of trust an individual may have with the robot they are interacting with. These factors include: Robot behaviour, Reliability, Level of automation, Proximity, Adaptability, Anthropomorphism, Communication, Task type, and what is harder to identify: human preference/behaviour. Within the domain that we are exploring, where a touch can determine the speed and state of the robot, there are several methods that can be used to develop a trust parameter to adjust for the robot's behaviour. Hanum et al. (2023) used motion capture to predict a user's confidence using the robot [6]. Loizaga et al. (2024) used psychophysiological signals, including brain activity, galvanic skin response, respiration, and pupillometry, to predict trust. Performance modelling and the use of Partially Observable Markov Decision Processes (POMDP) can also be used based on robot behaviours [7], [8]. These methods are all quite complex and require additional equipment and modelling for implementation, making them out of scope for this assignment. Thus, more algorithm-focused methods were identified to develop the trust parameter. Campagna et al. (2025) used a series of behavioural indicators between the robot and human (i.e., human attention to task, reaction time, legibility, human-robot speed synchronisation, predictability and spatial displacement) to train a preference-optimising based machine learning model [17]. The indicators used here were taken as inspiration for the expected model development for this system. For simplicity, we chose to focus on a linear model for the three variables: human-robot separation distance, human-robot speed synchronisation, and the reaction times.

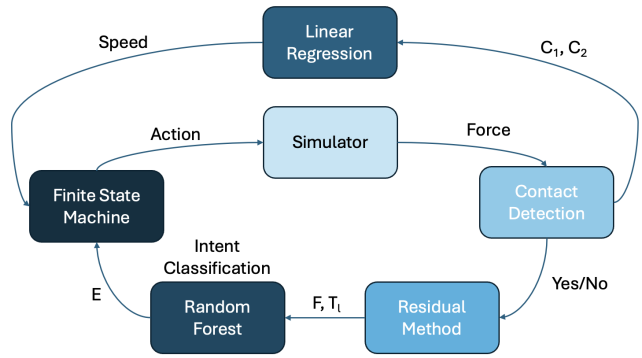

Fig. 2. A diagram of the full system to control the robot arm based on state and speed

III. Framework Development

Based on the research regarding human-robot collaboration, contact, and trust, we developed a framework that adjusts the robot's state and speed based on three inputs: force, trust parameter, and contact type. Figure 2 illustrates the full framework cycle and the relevant inputs added in each section. The framework is described in two phases based on the outputs: State and Speed.

A. State Output

The state to which the robot would move is determined by the force with which the participant touches the robot. A momentum observer [18] is used to estimate the discrepancy between commanded torque and model-based torques, referred to as the residual, which can be used to infer external contact forces. In practice, contact detection via the residual requires careful and often sensitive threshold tuning. As this work focuses on the Reaction phase of the collision event pipeline, we assume reliable contact detection and localization, and instead use a binary contact flag to simplify the framework. However, maintaining a well-defined residual is critical, as it provides a foundation for future work to implement contact detectioni through threshold tuning without redesigning the observer. The dynamic model during collision is modelled using the following.

$$M\ddot{q} + C\dot{q} + g = \tau + \tau_c \tag{1}$$

where q is the generalized coordinates, $\dot{q}$ the joint velocities, $M(q) > 0$ the inertia matrix, $C(q, \dot{q})$ the Coriolis/centrifugal terms, $g(q)$ the gravity vector, and τ the commanded torque. The joint torque τ_c associated to the contact force F_c is given by:

$$\tau_c = J_c^T F_c \tag{2}$$

where $J_c(q)$ is the Jacobian matrix of a frame Σ_c located at the contact point P.

Then, using the generalized momentum characterized by $p = M\ddot{q}$, the residual is modelled using the following:

$$r = K_I[p - \int_0^t (\tau + C^T\dot{q} - g + r)ds - p(0)] \tag{3}$$

Thus, the external force is estimated using the following:

$$F_c = \left(J_c^T\right)^+ \tau_c \approx \left(J_c^T\right)^+ r \tag{4}$$

where $\left(J_c^T\right)^+$ is the pseudoinverse of the contact Jacobian.

After the force is calculated based on the residual method, the start and end times of each touch are outputted to classify the contact type. This classification is done by a Random Forest Classifier. Random Forest Classifier is a type of classification learning method that computes several individual decision trees and outputs the class that has the highest vote of individual tree predictions [19]. It is known for its high accuracy and can help reduce overfitting that can be done by a single decision tree. Training is done on random subsets of generated data to create the individual trees. The respective forces at a given time and their start and end times are inputted to the Random Forest Algorithm. We used a maximum force of 15 N based on literature, which reviewed the magnitudes of forces that can be applied in different contexts [20]–[23]. While the context was not the same, the information helped to approximate the forces needed in this context. Based on the duration of each tap and the gap between contacts, the algorithm classifies the taps into three types: long tap, short tap, and double tap. Within the training data, a long tap is when the duration of the first force is greater than 3 ms and the duration of the second force is greater than 2 ms. A double tap is defined when the gap between two taps is 0.5; otherwise, it's a short tap. There is one more caveat: if there is no second force, then it can only be a short tap or a long tap. An overview of the tap classifications for the data is shown in Table I. This algorithm returns the classification as a 1x3 array with each cell representing a different form of tap.

TABLE I
TAP CLASSIFICATION USED IN DATA ANALYSIS

Contact Type	Definition
Long Tap	Duration of First Force <2
Double Tap	Time gap between two taps is <0.5
Short Tap	If above does not match

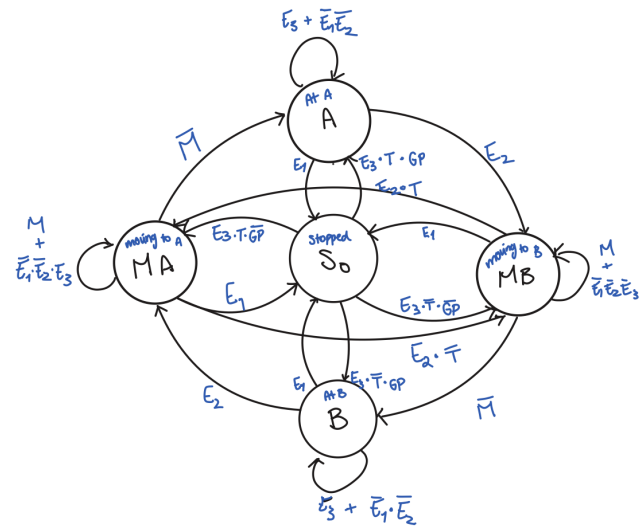

Fig. 3. A diagram of the finite state machine implemented in the simulation with appropriate actions and states defined

This event output is fed into a Finite State Machine (FSM), which dictates the state of the arm. The output is combined with knowledge of the states to compute the actions of the system. This Finite State Machine is shown in Figure 3, where it outlines all the events, states, and flags. The events E_1, E_2, and E_3 are the long tap, double tap, and single taps, respectively. The states include S_o (stopped state), MA (Moving to A State), MB (Moving to B state), A (location), and B (location). The binary flags include T (target is A), S (using the modulated speed), M (robot is in motion), and GP (robot has reached the goal position).

B. Speed Output

The speed output is determined by two parameters: trust and contact type. The trust factor is a value from 0 to 1. It was inspired by previous literature, which reviewed how trust can be incorporated into human-robot collaboration. Three inputs

TABLE II
FACTORS USED FOR THE PARAMETERS TO DETERMINE SPEED

Parameter	Factors Impacting
Trust	Distance Speed Synchronization Reaction Times
Contact	Contact Type

were chosen to comprise the trust parameter, C_1: human-robot separation distance, human-robot speed synchronization, and the reaction times. These three values would be averaged and then normalized between 0 and 1. The list of factors is outlined in Table II. For this study, we only completed the research and did not test actual normalization, and instead generated different values of trust between 0 and 1 for the algorithm. The contact type C_2 is a binary value, either 0 or 1. If a finger is used for contact, it is 1, and if a palm is used, this value is 0. Based on these two values, a linear regression equation was created to output the final speed of the robot using trust and contact type.

$$S = (\alpha C_1 + \beta C_2) S_0 \tag{5}$$

where S is the modified speed, S_0 is the maximum speed of the arm, and α and β are hyperparameters to be tuned. $\alpha > \beta$ and $\alpha C_1 + \beta C_2 < 1$.

IV. IMPLEMENTATION AND RESULTS

We implemented this system in the simulation software MuJoCo [1] on the Franka Emika Panda arm with a peg for an end effector. The code of the framework along with demonstration videos can be found in [24]. To test the system and the different cases, we ran various tests for different taps (short, long, double) on random points on the peg at different magnitudes. We purposefully did not include movements for the task at each box as we were interested in the behaviour of the arm moving to each location. Figure 4 showcases the model setup.

To test whether the simulation was working as expected, we ran several trials with three randomized touch events in each trial. The results of five trials are reported in Table III, comparing the true vs predicted label for the Random Forest, the expected vs actual action for the FSM, the RMSE of the residual, and the percent error of the contact force. While speed regulation qualitatively followed expected behaviour, it was not quantitatively evaluated in this work and may require further tuning and validation.

All 15 contact events were correctly classified by the Random Forest model and resulted in the appropriate FSM action, yielding 100% accuracy for both classification and control. While the residual RMSE and force error vary across events, the system consistently maintains correct decision-making and behaviour execution.

A summary of the results across all trials is shown in Table IV. The residual RMSE remains relatively consistent across all events, however the force error exhibits high variance mainly driven by a few outliers. Since the residual is accurate for those trials, future works could explore the validation of the contact Jacobian being used, according to Equation 4.

[1]https://mujoco.org/

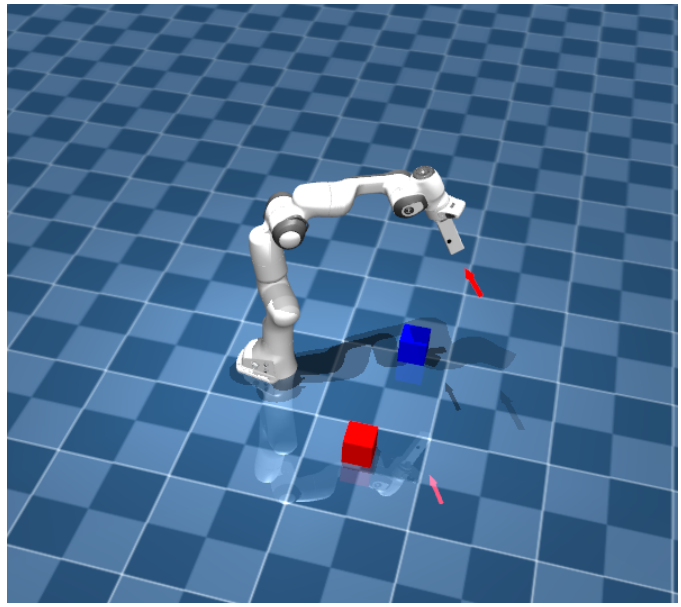

Fig. 4. The simulation of the framework with a modified Franka Panda Emika arm which has a peg at the end effector. The red and blue boxes represent the locations of A and B, respectively. The red arrow shows a scaled contact force with the point of application indicated by the black sphere on the peg of the robot.

TABLE III
EVALUATION RESULTS

Tr	**Ev**	**True**	**Pred**	**RF** (T/F)	**Exp**	**Act**	**FSM** (T/F)	**r** (RMSE)	F_c (Err %)
1	1	DT	DT	T	CT	CT	T	1.334	1.035
	2	DT	DT	T	CT	CT	T	0.692	13.261
	3	LT	LT	T	ST	ST	T	0.032	5.729
2	1	LT	LT	T	ST	ST	T	0.811	239.890
	2	DT	DT	T	CT	CT	T	0.604	2.843
	3	ST	ST	T	CN	CN	T	0.527	7.547
3	1	ST	ST	T	CN	CN	T	1.237	15.847
	2	LT	LT	T	ST	ST	T	0.457	6.682
	3	LT	LT	T	ST	ST	T	0.042	15.409
4	1	LT	LT	T	ST	ST	T	0.898	86.112
	2	DT	DT	T	CT	CT	T	0.205	7.644
	3	DT	DT	T	CT	CT	T	0.496	10.436
5	1	DT	DT	T	CT	CT	T	1.233	0.021
	2	ST	ST	T	CN	CN	T	0.741	7.195
	3	LT	LT	T	ST	ST	T	0.260	7.104

Note: DT: Double Tap, ST: Single Tap, LT: Long Tap; CT: Change Target, CN: Continue, ST: Stop.

V. USER STUDY

We propose the evaluation of this framework using an in-person study where participants will interact with the robot to complete a pick-and-place task. There will be two phases to the study: a) Robot Arm Practice and b) Robot Arm Task. We would recruit from the university population, and the total task time would be approximately 30-45 minutes.

TABLE IV
PER-TRIAL PERFORMANCE SUMMARY

Tr	RF (Acc %)	FSM (Succ %)	r (RMSE)	Force (Err %)
1	100	100	0.686 ± 0.651	6.67 ± 6.17
2	100	100	0.686 ± 0.651	6.67 ± 6.17
3	100	100	0.579 ± 0.606	12.65 ± 5.17
4	100	100	0.533 ± 0.348	34.73 ± 44.52
5	100	100	0.744 ± 0.487	4.77 ± 4.12
All	100	100	0.638 ± 0.480	28.45 ± 60.10

A. Research Questions

We explore the following research questions:

- **RQ1** Do participants trust the robot they are interacting with?
- **RQ2** Do participants find the robot easy to use?
- **RQ3** What aspects to change would make the participants feel more comfortable working with the robot?

We predict that participants who had parameters adjusted to fit their trust in the robot will trust the robot more and find it easier to use than those who do not have that personalisation.

B. Task Setup and Apparatus

We would use the Franka Emika Panda arm with two bins, where the arm has to move from one bin to the other to sort different colored foam blocks. The robot arm would be placed in the middle and the bins on each side so it is equi-distant.

C. Procedure

The study would involve participants completing a demographics survey prior to joining the study, and they would complete a sorting task with the robot arm.

Step 1 - Experience Questionnaire Before coming to the session, participants would complete a pre-survey questionnaire that would capture demographics (i.e., age, gender, faculty) and their past experiences with robots.

Step 2 - Robot Arm Practice Upon arrival at the study, participants would be instructed on the task where they have to guide the robot arm to pick up something in one box and then add it to another box. For the study, they will be using small foam blocks. The participant will need to go back and forth to pick and place 10 items. During this time, data will be collected to develop the trust parameter in the model. We will collect the parameters to inform the trust parameters in the algorithm: human-robot separation distance, human-robot speed synchronization, and the reaction times.

Step 3 - Robot Arm Task After they complete the first task, participants will be asked to complete the official task, where they will have to complete a sorting task to move small blocks of a certain colour into one box and move items of another colour into another box. The participant will either work with a robot arm that has been trained on their data or a general robot arm that has constant pieces added. We would collect task time, robot speed, and participant video footage for evaluation. After the experiment, participants would complete the Robotic Social Attributes Scale, System Usability Scale, and answer questions about what they liked when interacting with the robot and what they did not like when interacting. They would end by taking the Negative Attitude Towards Robots Survey.

VI. DISCUSSION AND FUTURE WORK

We were able to simulate our model in MuJoCo and demonstrate its potential as a possible way of using touch to adjust the robot's response in a human-robot collaboration setting. We tested different contact types in the system and simulated varying behaviours based on the constraints identified in the setup. All of the use cases worked as expected and lend themselves to a potential application in a real-world setup. While there are still a few cases that need adapting, with more focus on development, we anticipate these can be addressed. In particular, in certain trials the arm does not consistently maintain its position following contact. Additional refinement of the controller is required to ensure stable post-contact behaviour.

There are several areas of future work that we would like to explore moving forward. The first is the fine-tuning of the system parameters and optimizing for each of the algorithms used in the state and speed detection. There is an opportunity to optimize these parameters so there is a higher accuracy in the state selection. A sub-step may also be to test each section by itself (i.e., trust parameter creation, random forest algorithm), either empirically or in a user study, and understand if there are any adjustments.

The second area of improvement would be to implement this on an actual robot arm to test whether a participant would trust a personalized robot more and whether they would want to use it more due to the changes. With a formal user study, there will be an opportunity to understand whether this system works as intended and what further changes may be needed.

The final area of improvement would be to integrate the direction of the contact force to determine which target the robot should move to. Furthermore, the magnitude of the force can be integrated to further understand the role that force may play in the development of the trust parameter to manipulate the speed. While literature has not explored this extensively, we anticipate that the way in which the user interacts with the robot arm from a force perspective will provide some input into their trust in the robot arm as a whole.

VII. CONCLUSION

We proposed the design of a touch-based sensing system for collaborative robots that factors in trust to adjust the robot's behavior based on the user's preference. To do so, we use the joint positions to calculate a residual force of the robot. This force is used to identify the user intent, which is combined with a trust factor that predicts the speed at which the robot will move to the new requested state. The system is currently simulated on MuJoCo, where random forces are applied at random locations on the surface of the peg. Future work will include adjusting each of the parameters and subsystems for optimal performance, and then testing them in a physical

user study for verification. This system represents an initial design suggestion for touch-based human-robot collaboration that factors human trust

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